Notes: Jaffe, Trajtenberg & Henderson (QJE, 1993)

Geographic Localization of Knowledge Spillovers as Evidenced by Patent Citations

Contents

1 Big picture

- **Question.** Are knowledge spillovers geographically localized? If one inventor builds on another inventor's work, is the citing invention unusually likely to be near the original invention?
- **Empirical object.** The paper uses patent citations as observable traces of otherwise invisible knowledge flows. A citation from one patent to another is treated as evidence that the later invention drew on the earlier invention.
- **Core problem.** Innovative activity is geographically concentrated even without spillovers. To test localization, the paper compares actual citing patents to matched control patents from the same technology class and application year.
- **Main finding.** Patent citations are geographically more localized than matched controls at the country, state, and SMSA levels. The result is strongest for 1980 patents and remains visible after excluding self-citations.
- **Timing result.** Localization appears stronger for early citations and fades with citation lag, consistent with knowledge diffusing geographically over time.
- **Why self-citations matter.** Citations within the same assignee may reflect internal knowledge transfer rather than external spillovers. The paper therefore emphasizes non-self-citations.
- **Economic implication.** Knowledge has a spatial component. Even in patentable technology, spillovers are not fully global or frictionless; proximity matters for learning and invention.
- **Policy implication.** Geographic clusters, universities, and regional research ecosystems can generate local externalities that are not captured by the innovating firm or university.

2 Incremental contribution of the paper

- **Operationalizing knowledge spillovers.** The paper turns the abstract idea of spillovers into an empirical object by using patent citations as a paper trail of knowledge flows.
- **Matched-control design.** Prior evidence on agglomeration and innovation could be driven by the geographic concentration of industries. The paper creates matched control patents in the same patent class and application year, making the benchmark geographically comparable.
- **Multiple levels of geography.** The paper tests localization at the country, state, and SMSA levels. This separates broad domestic concentration from genuinely local spatial spillovers.
- **Self-citation discipline.** By distinguishing self-citations from external citations, the paper separates internal firm/university knowledge transfer from potential external spillovers.
- **Dynamic spillover evidence.** The citation-lag analysis adds a time dimension. Localization is not just a static fact; it appears to fade as knowledge diffuses.
- **Foundation for later patent-citation research.** The paper helped establish patent citations as a central empirical tool for studying innovation, knowledge diffusion, and geography.

Interpretation: The incremental contribution is not simply finding that citations are local. It is constructing a credible counterfactual: where would a citation have come from if it reflected only the geographic distribution of related inventive activity rather than knowledge spillovers from the origin patent?

3 Data and measurement

3.1 Patent cohorts

The paper starts from patents granted in two application cohorts, 1975 and 1980. It studies citations to these originating patents observed through 1989. Originating patents are grouped into three categories:

- university patents;
- top corporate patents;
- other corporate patents.

Interpretation: The comparison across universities and corporations is important because universities may diffuse knowledge through academic channels, while firms may rely more on internal or local business channels.

3.2 Patent citations as knowledge flows

A citation from a later patent to an earlier patent is interpreted as a possible knowledge spillover. The citation lag is defined as:

$$\text{Citation lag} = \text{application year of citing patent} - \text{application year of originating patent}.$$

Interpretation: Citations are noisy evidence. Some citations are inserted by patent examiners, some by applicants, and some may reflect legal rather than technological knowledge. The matched-control design is meant to recover a statistical pattern despite this noise.

3.3 Self-citations

A self-citation is a citation where the citing patent and the originating patent are assigned to the same party. The paper reports results both with and without self-citations.

Interpretation: Self-citations may reveal successful internal appropriation of knowledge, but they are less clean as evidence of external spillovers. Excluding them makes the test more conservative.

3.4 Geographic units

The paper measures whether originating and citing patents match geographically at three levels:

- same country;
- same U.S. state;
- same SMSA.

Interpretation: The SMSA match is the strongest test of local spillovers. A country match may reflect national patenting patterns, but an SMSA match is much closer to the idea of localized knowledge flow.

3.5 Matched controls

For each actual citation, the paper constructs a control patent matched by application year and three-digit technology class. The control patent is not the citing patent; it is a benchmark patent that represents the geographic distribution of similar inventive activity.

Interpretation: This is the key measurement step. If the actual citation is more geographically proximate to the origin patent than the matched control, the excess proximity is evidence of localization beyond industry geography.

4 Models and empirical specifications

4.1 Raw matching fractions

The first empirical object is a matching rate:

$$Match^{geo} = \Pr\{\text{citing patent and originating patent are in same geography}\}.$$

For the control sample:

$$ControlMatch^{geo} = \Pr\{\text{matched control patent and originating patent are in same geography}\}.$$

The localization test compares:

$$Match^{geo}_{citations,\ excl.\ self} - Match^{geo}_{controls}.$$

Interpretation: A positive difference means that citations are more local than one would expect from the location of technologically similar patents.

4.2 Difference-in-proportions test

Table III reports t-statistics for equality of the citation matching proportion excluding self-cites and the control matching proportion.

$$H_0 : Match^{geo}_{citations,\ excl.\ self} = Match^{geo}_{controls}.$$

Interpretation: This is the simplest test of localization. It does not yet adjust for citation lag or other patent-level controls.

4.3 Geographic probit model

The probit model pools citations excluding self-cites and estimates the probability of a geographic match:

$$Pr(Match^{geo}_i = 1) = \Phi\left(\alpha + \beta_1 ControlMatch_i + \beta_2 \log(Lag_i) + \beta_3 TopCorp_i + \beta_4 OtherCorp_i + \Gamma' X_i\right).$$

Controls include interactions of lag with corporate-origin indicators, same patent-class match, generality of the origin patent, origin self-citation fraction, and technology-field dummies.

Interpretation: The probit model asks whether localization fades with lag and whether patent or origin characteristics explain the probability of a local citation.

4.4 Predicted localization over time

The probit estimates imply predicted localization probabilities at different lags. Table V evaluates these probabilities using the 1975 university-patent lag coefficient:

$$\widehat{Pr}(Same\ Geography \mid Lag = t, X = \bar{X}).$$

Interpretation: This transforms probit coefficients into economically meaningful probabilities. It shows how local spillovers fade over five, ten, and twenty-five years.

5 Identification and empirical design

5.1 Core identification logic

The empirical challenge is that innovative activity is geographically clustered. A later patent may be near an earlier patent not because of a spillover, but because both are in the same industrial region. The matched-control design solves this by comparing each actual citation with a patent that shares the same technology class and application year.

$$\text{Localization evidence} = \text{actual citation proximity} - \text{matched-control proximity}.$$

Interpretation: The control patent is the counterfactual. It asks: how geographically close would a technologically similar patent have been absent a citation link?

5.2 Why patent-class and year matching matter

Patent class matching controls for technology geography. Year matching controls for changes in the geographic distribution of patenting over time and for the fact that the origin patents are from particular application cohorts.

Interpretation: Without these controls, evidence of localization could be an artifact of the semiconductor industry being in Silicon Valley or medical technology being near Boston.

5.3 Why excluding self-cites matters

Self-citations can arise because an organization builds on its own work. That is real knowledge flow, but not necessarily an external spillover. Excluding self-cites makes the localization evidence more directly about knowledge moving across assignees.

Interpretation: The fact that localization persists after excluding self-cites is one of the strongest pieces of evidence in the paper.

5.4 Why citation lag matters

If proximity matters because face-to-face interaction, local labor mobility, or local professional networks help transmit knowledge, localization should be strongest soon after the origin invention and fade as knowledge diffuses more broadly.

Interpretation: The lag pattern is a mechanism test. Fading over time supports the idea that localization reflects temporary diffusion frictions rather than only fixed local industrial structure.

5.5 Threats to interpretation

- **Citation noise.** Patent citations may be added by examiners and may not always reflect true knowledge flows.

- **Control mismatch.** A three-digit patent class may not perfectly capture technological relatedness.
- **Inventor mobility.** Local citations may partly reflect inventor movement rather than pure spillovers.
- **Organization networks.** Non-self-citations could still occur within long-term supplier, customer, or alliance networks.
- **Administrative geography.** SMSA and state boundaries are imperfect measures of actual knowledge networks.

Interpretation: These limitations mean the results should be interpreted as statistical evidence of localization, not as a direct map of every knowledge path.

6 Tables and figures

6.1 Table I: Descriptive Statistics; Figure I: Citation lag distributions

Read:

- Table I shows that most origin patents receive at least one citation: roughly 74–89 percent depending on cohort and origin type.
- University patents receive more citations on average than corporate patents, especially in the 1975 cohort.
- Self-citations are much lower for universities and higher for top corporations.
- Figure I shows that citation counts rise after application and plateau after about three years.
- The 1980 cohort has a shorter observed tail because citations are only observed through 1989.

Economic message: Table I establishes the citation sample and the importance of excluding self-cites. Figure I shows why citation timing must be considered when studying localization.

Detailed interpretation: University patents appear to be cited more broadly and less internally, which is consistent with universities producing more open or publicly diffused knowledge. The lag distribution also shows that citations are not instantaneous; there is a diffusion process.

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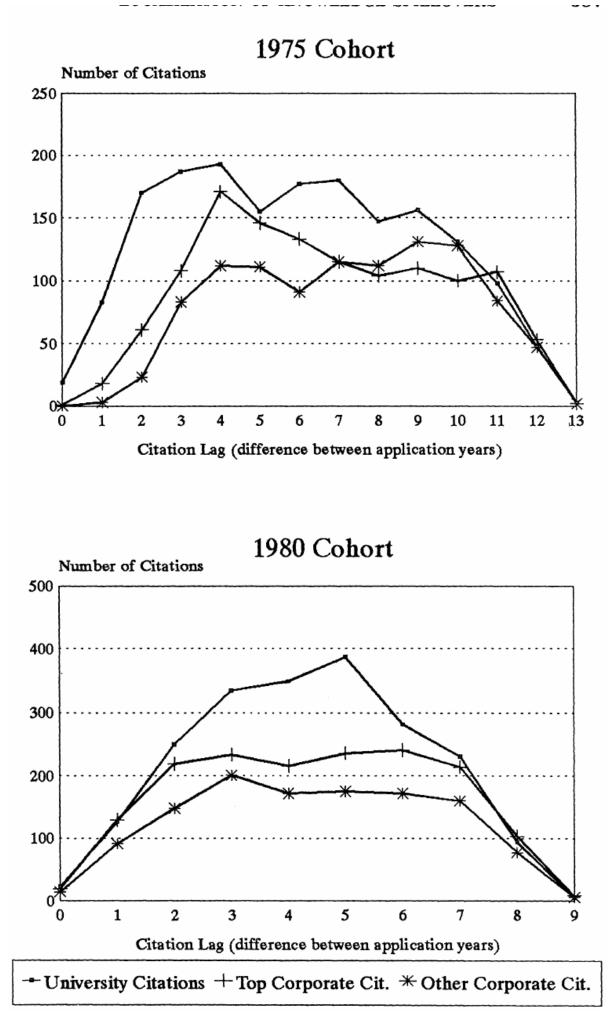
TABLE I
DESCRIPTIVE STATISTICS

Originating dataset	Percent receiving citations	Total no. of citations	Mean citations received	Average citation lag[a,b]	Percent self-citations[b]	Percent same patent class[b,c]
1975						
University	88.6	1933	6.12	6.53	5.6	54.3
Top corporate	84.2	1476	4.70	7.17	18.6	55.7
Other corporate	82.3	1341	4.22	7.82	9.1	57.5
1980						
University	79.9	2093	4.34	4.36	8.9	56.3
Top corporate	79.9	1701	3.54	4.41	24.6	58.3
Other corporate	74.1	1424	2.95	4.46	12.6	57.2

a. Application year of citing patent minus application year of originating patent.
b. For those patents receiving any citations.
c. Comparison is at the three-digit level (see text).

(a) **Table I: Descriptive Statistics.** Citation rates, citation counts, citation lags, self-citation rates, and same patent-class rates by cohort and origin type.



(b) **Figure I: Citation lag distributions.** Citation counts by lag for university, top corporate, and other corporate patents.

Descriptive evidence on citations and timing. These visuals establish the citation universe and show that lag structure matters for later localization tests.

6.2 Table II: SMSA Distributions for Some Datasets; Table III: Geographic Matching Fractions

Read:

- Table II shows that patenting and citing activity is geographically concentrated, especially in major SMSAs such as Boston, Los Angeles, San Francisco, and New York.
- It also shows why controls are necessary: technologically related patents are not randomly distributed across geography.
- Table III compares actual citation matches with matched-control matches at the country, state, and SMSA levels.
- Excluding self-cites, citations are much more local than controls, especially for 1980 patents.
- At the SMSA level, 1980 non-self citations are roughly two to six times as likely to match as controls, depending on origin type.

Economic message: Table II motivates the control design; Table III is the paper's main raw localization evidence.

Detailed interpretation: The most important comparison is not overall geographic concentration. It is the difference between actual citations excluding self-cites and control patents. For example, actual SMSA matching rates excluding self-cites in 1980 range from 6.9 to 8.8 percent, while controls range from 1.1 to 3.6 percent.

TABLE II
SMSA DISTRIBUTIONS FOR SOME DATASETS

Location	1975 University originating	1975 Top corporate originating	Citations to 1975 university	Citations to 1975 top corporate	Controls for citations to 1975 university	All citations to patents from in NY SMSA
Foreign	—	—	31.8	31.4	35.8	31.2
Boston	15.0	3.1	7.5	4.6	5.1	4.0
Los Angeles/ Anaheim	7.0	4.8	9.0	5.7	6.1	3.9
San Francisco/ Oakland	5.1	1.4	3.8	3.7	6.1	3.5
Madison, WI	4.2	—	1.6	—	0.5	0.6
Philadelphia/ Wilmington	4.2	9.3	5.4	8.2	4.5	9.1
Rural Iowa	3.8	—	1.6	0.6	0.2	—
San Jose	3.5	2.8	4.0	3.4	—	1.9
New York/ NJ/CT	3.2	13.5	9.7	11.7	13.7	28.5
Salt Lake City	3.2	—	2.1	—	0.5	0.4
Detroit/ Ann Arbor	2.6	2.4	2.6	1.7	1.7	1.2
Minneapolis/ St. Paul	1.3	5.2	2.8	2.9	1.9	2.1
Chicago	1.9	4.2	3.9	5.7	5.6	4.2
Albany	0.6	3.1	1.9	2.1	1.3	0.8

All figures are percentages. SMSA percentages for citations and controls are relative to *domestic* total.

(a) **Table II: SMSA Distributions for Some Datasets.** Geographic concentration of origin patents, citations, controls, and New York-origin citations.

TABLE III
GEOGRAPHIC MATCHING FRACTIONS

	1975 Originating cohort			1980 Originating cohort		
	University	Top corporate	Other corporate	University	Top corporate	Other corporate
Number of citations	1759	1235	1050	2046	1614	1210
		Matching by country				
Overall citation matching percentage	68.3	68.7	71.7	71.4	74.6	73.0
Citations excluding self-cites	66.5	62.9	69.5	69.3	68.9	70.4
Controls	62.8	63.1	66.3	58.5	60.0	59.6
t-statistic	2.28	−0.1	1.61	7.24	5.31	5.59
		Matching by state				
Overall citation matching percentage	10.4	18.9	15.4	16.3	27.3	18.4
Citations excluding self-cites	6.0	6.8	10.7	10.5	13.6	11.3
Controls	2.9	6.8	6.4	4.1	7.0	5.2
t-statistic	4.55	0.09	3.50	7.90	6.28	5.51
		Matching by SMSA				
Overall citation matching percentage	8.6	16.9	13.3	12.6	21.9	14.3
Citations excluding self-cites	4.3	4.5	8.7	6.9	8.8	7.0
Controls	1.0	1.3	1.2	1.1	3.6	2.3
t-statistic	6.43	4.80	8.24	9.57	6.28	5.52

Number of citations is less than in Table I because of missing geographic data for some patents. The

(b) **Table III: Geographic Matching Fractions.** Actual citation localization compared with matched-control localization.

Controls and matching fractions. Pairing the tables shows why matched controls are necessary and how actual citations exceed that benchmark.

6.3 Table IV: Geographic Probit Results; Table V: Predicted Localization Percentages over Time

Read:

- Table IV estimates probit models for same-country, same-state, and same-SMSA matches in the 1975 and 1980 cohorts.
- The control-match dummy absorbs baseline geographic concentration in related patenting.
- Citation lag is negative for university patents in the 1975 state and SMSA regressions, consistent with fading localization.
- Origin self-citation fraction is positively related to localization in state and SMSA models.
- Table V translates the 1975 university lag coefficient into predicted localization rates over time.

Economic message: The probit results show that localization is not only a raw matching fact. It remains meaningful after controlling for control-patent geography, technology, lag, origin type, and origin-patent characteristics.

Detailed interpretation: Table V makes the fading mechanism easier to read: predicted same-state localization for university citations falls from 9.7 percent at lag 0–1 to 5.3 percent at ten years and 4.0 percent at twenty-five years. SMSA localization declines from 4.8 to 3.3 percent.

TABLE IV
GEOGRAPHIC PROBIT RESULTS

	Country match		State match		SMSA match	
	1975	1980	1975	1980	1975	1980
Dummy for control	0.139	0.085	0.396	0.300	*	0.283
Sample match	(0.045)	(0.041)	(0.124)	(0.102)		(0.172)
Log of citation	−0.078	0.094	−0.264	0.198	−0.123	0.037
lag	(0.049)	(0.056)	(0.073)	(0.079)	(0.057)	(0.086)
Dummy for top	−0.114	−0.010	−0.383	0.013	−0.234	−0.208
corporate	(0.168)	(0.127)	(0.249)	(0.177)	(0.288)	(0.200)
Dummy for other	0.069	0.053	−0.214	−0.007	0.325	−0.042
corporate	(0.209)	(0.134)	(0.277)	(0.189)	(0.291)	(0.207)
Log-lag	0.046	−0.016	0.226	0.007	0.102	0.156
*top corp. dummy	(0.091)	(0.086)	(0.138)	(0.115)	(0.156)	(0.131)
Log-lag	0.008	−0.026	0.307	0.036	0.037	0.039
*other corp. dummy	(0.108)	(0.091)	(0.147)	(0.124)	(0.155)	(0.138)
Dummy for matching	−0.085	0.069	−0.013	0.034	−0.057	−0.016
patent class	(0.050)	(0.045)	(0.073)	(0.058)	(0.080)	(0.068)
Generality of origin	0.092	0.177	0.026	−0.140	0.013	−0.298
patent	(0.091)	(0.088)	(0.136)	(0.111)	(0.150)	(0.130)
Origin fraction	−0.813	0.162	0.815	0.883	1.174	0.828
Self-citations	(0.180)	(0.124)	(0.246)	(0.134)	(0.237)	(0.154)
# of observations	3581	4217	3573	4215	3566	3972
# of matches	2363	2925	256	490	197	298
Log likelihood	−2269	−2559	−894	−1459	−736	−1022

*The number of observations for which the control patent matched at the SMSA level was so small that this parameter could not be estimated.
Standard errors are in parentheses. All equations also included five technological field dummies.

(a) **Table IV: Geographic Probit Results.** Probit estimates for country, state, and SMSA matching.

TABLE V
PREDICTED LOCALIZATION PERCENTAGES OVER TIME (BASED ON 1975 PROBIT RESULTS FOR CITATIONS OF UNIVERSITY PATENTS)

	Predicted percentage for:		
Lag	Same country	Same state	Same SMSA
0 or 1 year	67.1	9.7	4.8
5 years	65.5	6.5	4.0
10 years	64.6	5.3	3.7
25 years	63.5	4.0	3.3

(b) **Table V: Predicted Localization Percentages over Time.** Predicted decline in localization as citation lag grows.

Adjusted localization and fading over time. Table IV estimates the multivariate model; Table V converts lag coefficients into predicted probabilities.

7 Short summary

- The paper tests whether knowledge spillovers are geographically localized using patent citations.
- Patent citations are treated as traces of knowledge flows from earlier inventions to later inventions.
- The core identification strategy compares actual citations with matched control patents in the same patent class and application year.
- Citations are more geographically localized than controls at country, state, and SMSA levels.
- The result is strongest for 1980 patents and remains after excluding self-citations.
- University patents receive more citations and fewer self-citations than corporate patents.
- Localization appears to fade as citation lag increases.
- Probit models confirm the localization pattern after controlling for control-patent geography, technology, origin type, lag, generality, and self-citation intensity.

- The paper helped establish patent citations as a way to study knowledge diffusion and geography.

8 Conclusion

8.1 Core conclusion

Jaffe, Trajtenberg, and Henderson provide evidence that knowledge spillovers are geographically localized. Later patents that cite an earlier patent are more likely to come from the same country, state, or SMSA than matched control patents in the same technology class and application year. This difference is especially strong at the SMSA level and remains after excluding self-citations.

8.2 Economic intuition

Knowledge does not move frictionlessly. Inventors learn through local networks, face-to-face interaction, labor mobility, seminars, supplier-customer relationships, and informal observation. Patent citations capture only one trace of these flows, but their geographic pattern reveals that proximity matters.

8.3 Incremental contribution revisited

The paper's key contribution is the counterfactual benchmark. Without matched controls, localized citations could simply reflect the fact that related technology is clustered geographically. By matching control patents on technology class and application year, the paper asks whether citations are more local than the underlying geography of invention. That design created a foundation for later empirical work on spillovers, patent citations, and regional innovation.

8.4 Policy implications

If knowledge spillovers are local, then universities, research labs, and innovative firms may create local externalities. Regional innovation policy, university research support, and cluster formation can have effects beyond the direct output of the funded institution. The results also suggest that physical proximity remains important even in codified technological knowledge.

8.5 Limitations

Patent citations are imperfect measures of knowledge flows. Some are added by examiners, some may be legal rather than technological, and some spillovers leave no citation. Geographic units such as SMSAs are also imperfect proxies for actual social networks. The paper is therefore best read as statistical evidence of localization, not a direct observation of all knowledge transfers.

8.6 Takeaway

The main takeaway is that ideas leave geographic footprints. Patent citations are more localized than comparable inventive activity, and this localization weakens over time. Knowledge spillovers are partly local, especially soon after invention.